

Builder Journal — Week 1

Why did you choose Computer Vision?

I chose Computer Vision because the visual, hands-on nature of the field genuinely interests me — I find myself far more drawn to problems I can see and interact with than to purely abstract ones. I also believe Computer Vision has strong long-term relevance, with rapid global adoption across industries and a particularly close tie to robotics. My final year project reinforced this interest directly: a smart mirror that analyzes a person's face shape in real time and suggests a matching hairstyle and beard style, then overlays the suggestion live on the mirror feed. Working on that project is what first pulled my interest toward Computer Vision specifically.

What applications interest you most?

My primary areas of interest are autonomous vehicles, robotics, and AR/VR. Robotics stands out to me as both technically advanced and high in future scope, making it a field I want to build deeper expertise in. AR/VR is directly relevant to my final year project, so pursuing it further lets me build on experience I've already started gaining rather than starting from zero.

What did you learn this week?

Technical learnings:

I learned how to preprocess images using OpenCV and applied this in a real, functioning web application rather than isolated scripts. This was also my first time building a complete web app, which meant learning HTML, CSS, JavaScript, and FastAPI, and understanding how each fits into the overall system. On the computer vision side, I learned the distinction between image classification and object detection, how manual data labeling is done, and the difference between contour detection and edge detection.

Non-technical learnings:

I learned how to set up a GitHub repository in a professional, structured way, and how to properly document a system's architecture. Beyond that, one of the most useful takeaways was understanding the full set of steps required to take a finished project from documentation through to deployment.

Biggest challenge?

My biggest challenge was building the web application itself. I was already comfortable with OpenCV and had no trouble implementing the core computer vision functions, but I had never built a backend before, and I had no prior experience with frontend development using HTML/CSS either. Connecting the two — frontend and backend — was the part I struggled with most.

How did you solve it?

I worked through this challenge by studying YouTube tutorials and using AI tools to help me understand the concepts I was missing. I initially generated a working version of the app using Cursor AI, but I made a deliberate choice to rebuild it from scratch myself instead of relying on that output, using AI assistance only to help me understand HTML, CSS, and FastAPI as I coded. By the end of the week I could explain how the frontend and backend integrate and why FastAPI is a useful framework for this purpose. I also attempted to deploy the application

publicly using Railway and Render, but ran into deployment issues I wasn't able to resolve by week's end — this remains unfinished and is a carryover into next week.

Goals for Week 2:

My goals for Week 2 are to successfully deploy this web application and make it publicly accessible, resolving the deployment issues from this week, and to build an additional application focused on object detection.